NOVA-RF1

Hand-held multi-band RF / NFC / sub-GHz / IR multi-tool — Preliminary Datasheet, Rev v0.1

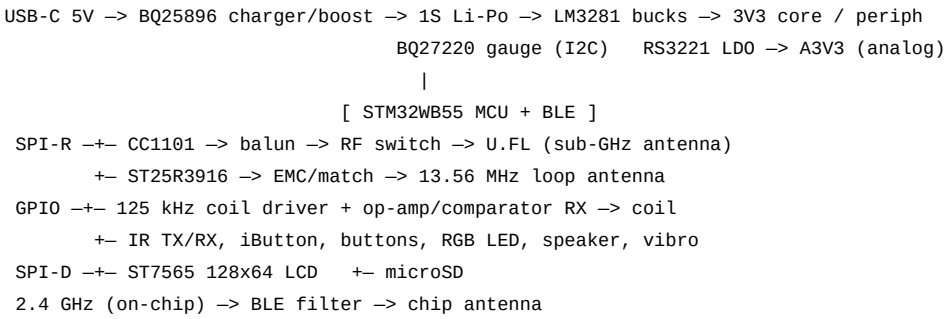
Features	Applications
STM32WB55RGV6: Cortex-M4 64 MHz + Cortex-M0+ BLE 5.0, 1 MB flash, 256 KB RAM	RFID / NFC development & research
Sub-GHz CC1101: 300-348 / 387-464 / 779-928 MHz, OOK/2-FSK/GFSK/MSK, up to +12 dBm	Sub-GHz remote capture & replay
13.56 MHz NFC ST25R3916: reader / card-emulation / peer, ISO14443A/B, 15693, FeliCa	Access-control / CTF / education
125 kHz RFID discrete AFE (read EM4100 / HID-class)	IR universal remote
IR TX (38 kHz) + learning RX; 1-Wire / iButton	1-Wire / iButton tooling
128x64 mono LCD, microSD, RGB LED, speaker, vibro, 6-key UI	BLE peripheral / beacon work
USB-C charge (BQ25896) + fuel gauge (BQ27220), 1S Li-Po	Embedded / hardware hacking
18-pin expansion header (GPIO, SPI, UART, I2C, SWD, 3V3/5V)	Open firmware (Flipper FW compatible)

General description

NOVA-RF1 is an original open-hardware hand-held multi-tool that integrates five radio/signalling front-ends around an STM32WB55 wireless MCU. A single Cortex-M4 application core runs the UI and protocol logic while an on-chip Cortex-M0+ handles the BLE 5.0 stack. Peripherals attach over two SPI buses, an I²C power bus, and timer/GPIO lines following the pin map of the open-source Flipper Zero firmware (GPLv3), which the board is designed to run. The board is 100 × 60 mm, 4-layer, two-sided SMT, in a two-part hand-held enclosure.

This is an original design: its schematic, PCB, outline, silkscreen and name are its own; it uses none of Flipper Devices' trademarks, logo, dolphin mascot or hardware design files. The CC1101 sub-GHz transmitter and the BLE radio require RF type approval (FCC / CE / WPC-ETA + BIS) before a product is sold or operated; transmit only on licensed bands and powers.

Block diagram



Bus & interface map

Bus	MCU pins	Devices
SPI-R (radio)	SCK PA5 / MISO PB4 / MOSI PB5	CC1101 (CS PD0, GDO0 PA1); ST25R3916 (CS PE4)
SPI-D (display)	SCK PD1 / MISO PC2 / MOSI PB15	ST7565 LCD (CS PC11, DC PB1, RST PB0); microSD (CS PC12, CD PC10)
I2C power	SDA PA10 / SCL PA9	BQ25896 charger, BQ27220 gauge, LP5562 LED driver
USB-FS	DM PA11 / DP PA12	USB-C (via USBLC6 ESD)
SWD	SWDIO PA13 / SWCLK PA14 / NRST	Tag-Connect TC2030
UART1	TX PB6 / RX PB7	expansion header

Expansion header J_GPIO (2×9, 2.54 mm)

Pin	Net	Pin	Net
1	+3V3	2	GPIO PA7
3	GPIO PA6	4	GPIO PA4
5	GPIO PB3	6	GPIO PB2
7	GPIO PC3	8	GPIO PC1
9	GPIO PC0	10	SWCLK
11	SWDIO	12	UART TX (PB6)
13	UART RX (PB7)	14	I2C SDA (PA10)
15	I2C SCL (PA9)	16	VBUS (+5V)
17	VIBRO (PA8)	18	GND

Other connectors

Ref	Type	Function
J_USB	USB-C receptacle	5 V charge input + USB-FS data
X10	1.25 mm 3-pin	external 1S Li-Po: BAT / NTC / GND
X6	18-pin 0.5 mm FFC	link to NFC/RFID/IR sub-section (if partitioned)
J_SUBG	U.FL	sub-GHz antenna
J_SWD	Tag-Connect TC2030	SWD programming / debug

Functional description

MCU & BLE (U1, STM32WB55RGV6)

Dual-core wireless MCU: a Cortex-M4 at up to 64 MHz runs the application, a Cortex-M0+ runs the certified BLE 5.0 stack. 1 MB flash / 256 KB RAM, USB-FS, AES/PKA crypto. A 32 MHz HSE crystal (Y1, under the chip) clocks the radio and an internal SMPS (L_SMPS 10 uH) powers it; a 32.768 kHz LSE (Y2) clocks the RTC. The 2.4 GHz BLE output passes through a band-pass filter (FLT1) and pi-match to a chip antenna (ANT1).

Sub-GHz transceiver (U2, CC1101)

Configured over SPI-R. Supports 300-348, 387-464 and 779-928 MHz with OOK, 2-FSK, GFSK and MSK, data rates to 600 kbps, sensitivity to ~-112 dBm and output to ~+12 dBm. The differential RF port is converted by a balun (FLT2) and LC match, then two SPDT RF switches (SW1/SW2) select band/antenna to the U.FL output. A 26 MHz crystal (ZQ3) sets the reference; RBIAS = 56 k.

13.56 MHz NFC (U3, ST25R3916)

Reader / card-emulation / active-peer front-end on SPI-R, clocked by a 27.12 MHz crystal (ZQ1). The differential driver feeds an EMC + matching network (L1/L2 270 nH plus the tuning caps) into a PCB loop antenna; the receiver reads card modulation on RFI_P/RFI_N. Supports ISO14443A/B, ISO15693 and FeliCa.

125 kHz RFID analog front-end

A FET H-bridge (VT4/VT6) drives the read coil with a 125 kHz carrier gated by an MCU timer. The returning amplitude-modulated signal is band-pass amplified by two op-amps (DA3/DA4, LMV358) on the clean A3V3 rail, sliced by a comparator (DA5, LMV331) against a 1.65 V mid-rail (R_MID divider), and read back as digital data. Reads EM4100 / HID-class tags.

Infrared, 1-Wire, UI

IR: a VSMY14940 LED bank driven from a timer pin transmits 38 kHz-modulated codes; a TSOP75338 demodulator receives for learning. iButton: an ESD-protected open-drain GPIO (PB14). UI: 128x64 ST7565 LCD and microSD on SPI-D, a 6-key cluster, an RGB LED (LP5562), a magnetic speaker and a vibromotor.

Power

USB-C 5 V feeds a BQ25896 1-cell Li-ion charger with 5 V OTG boost and I2C control; a BQ27220 fuel gauge reports state-of-charge. Two LM3281 bucks generate the 3.3 V core and peripheral rails; an RS3221 LDO supplies the clean analog rail (A3V3); a TPS22917 load switch cuts the peripheral rail in deep sleep. The battery is external (X10).

Absolute maximum ratings

Parameter	Min	Max	Unit
USB-C VBUS input	-0.3	6.0	V
Battery (VBAT)	-0.3	4.5	V
Logic I/O (3V3 domain)	-0.3	3.6	V
Operating temperature	-20	+60	degC
Storage temperature	-40	+85	degC

Recommended operating conditions

Rail	Typ	Range	Source / load
VBUS	5.0 V	4.4-5.25 V	USB-C charge in / OTG out
VBAT	3.7 V	3.0-4.2 V	1S Li-Po
+3V3 (core)	3.3 V	+/-3%	LM3281 buck - MCU, radios, display
+3V3_PERIPH	3.3 V	+/-3%	switched (load switch) - SD/radios
A3V3 (analog)	3.3 V	+/-2%	RS3221 LDO - RFID op-amps

Estimated current (indicative)

Mode	Typ current (at VBAT)
Idle (display on, radios off)	~25-40 mA
Sub-GHz RX / NFC poll	~40-70 mA
Sub-GHz TX (+12 dBm) burst	~120-160 mA peak
Deep sleep (peripheral rail off)	~100-300 uA

Current figures are design estimates for budgeting; confirm on hardware.

RF summary

Radio	Band	Notes
Sub-GHz (CC1101)	300-928 MHz (ISM sub-bands)	OOK/FSK/GFSK/MSK; antenna via U.FL + RF switch
NFC (ST25R3916)	13.56 MHz	ISO14443A/B, 15693, FeliCa; PCB loop antenna
125 kHz RFID	125 kHz	EM4100 / HID-class read; wound/PCB coil
BLE (STM32WB55)	2.400-2.4835 GHz	BLE 5.0; chip antenna via band filter

Mechanical

Parameter	Value
Board size	100 x 60 mm, rounded handheld outline
Layers / thickness	4-layer (Sig/GND/PWR/Sig), 1.6 mm, ENIG
Enclosure	two-part printed case, outer ~106 x 66 mm
Assembly	double-sided SMT (ICs/connectors top, passives bottom)

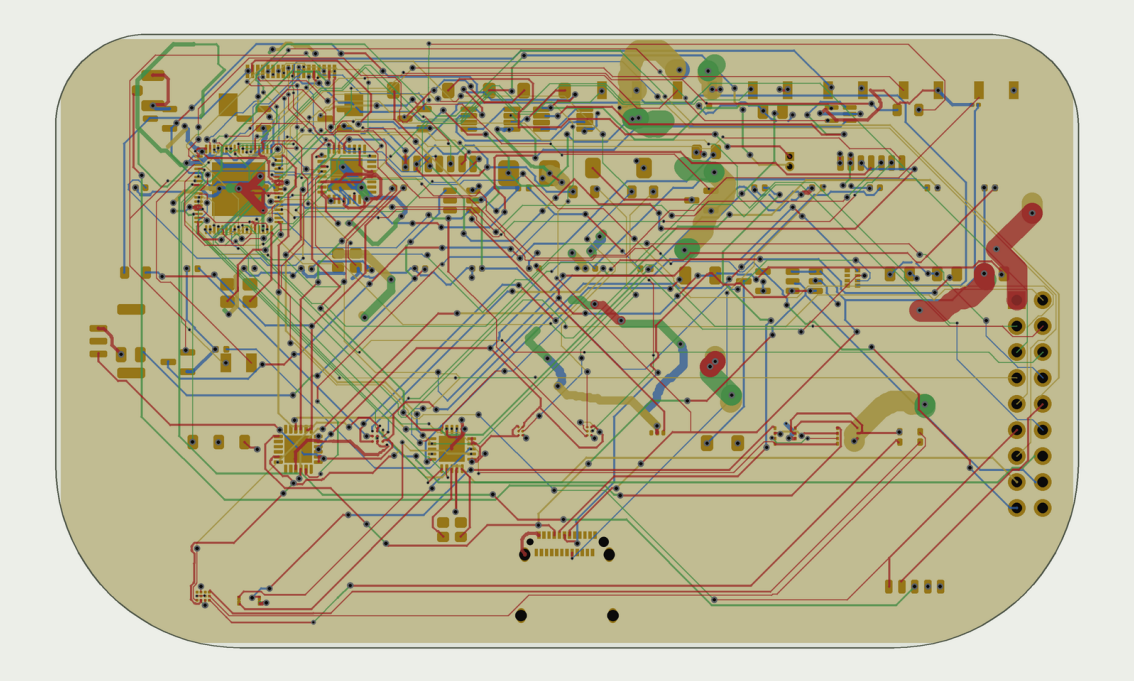


Figure: routed 4-layer board (red=top, blue=bottom, inner GND/PWR planes, dots=vias).

Ordering: NOVA-RF1 v0.1, 4-layer PCBA, original open-hardware. Compliance: radios require type approval (FCC/CE/WPC-ETA+BIS) before sale or transmit. Preliminary - all values subject to change and hardware verification. Original design; not a Flipper Zero; no Flipper Devices IP used.